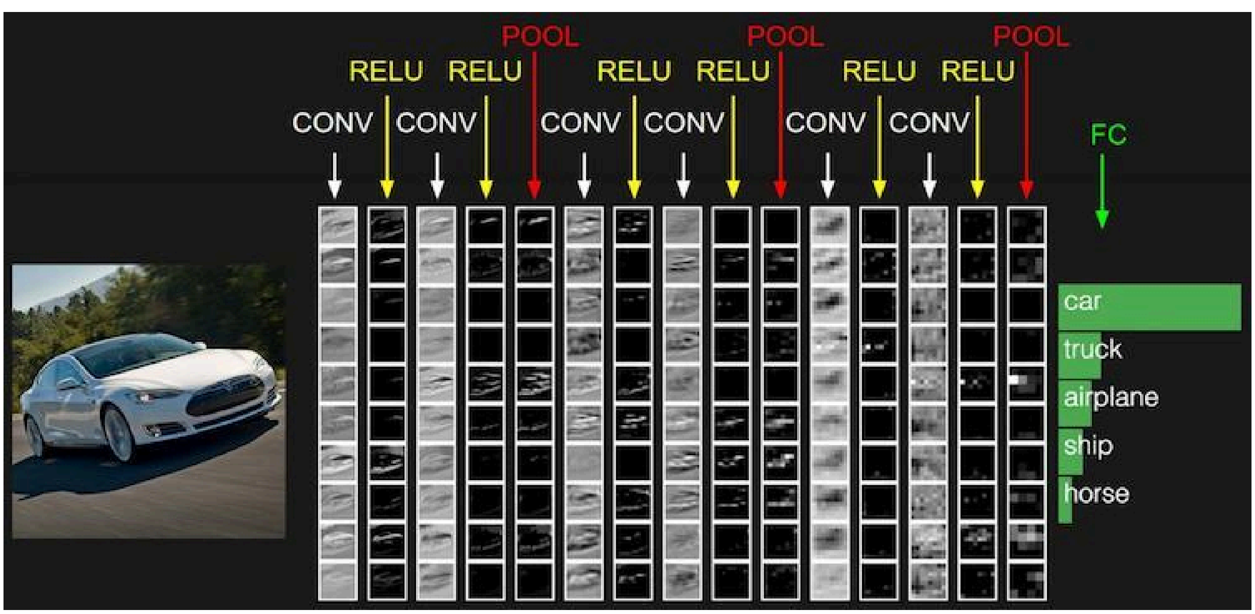
Computer Vision Engineer Roadmap

Technologies: OpenCV + CNN + YOLO

OpenCV

python




YOLO

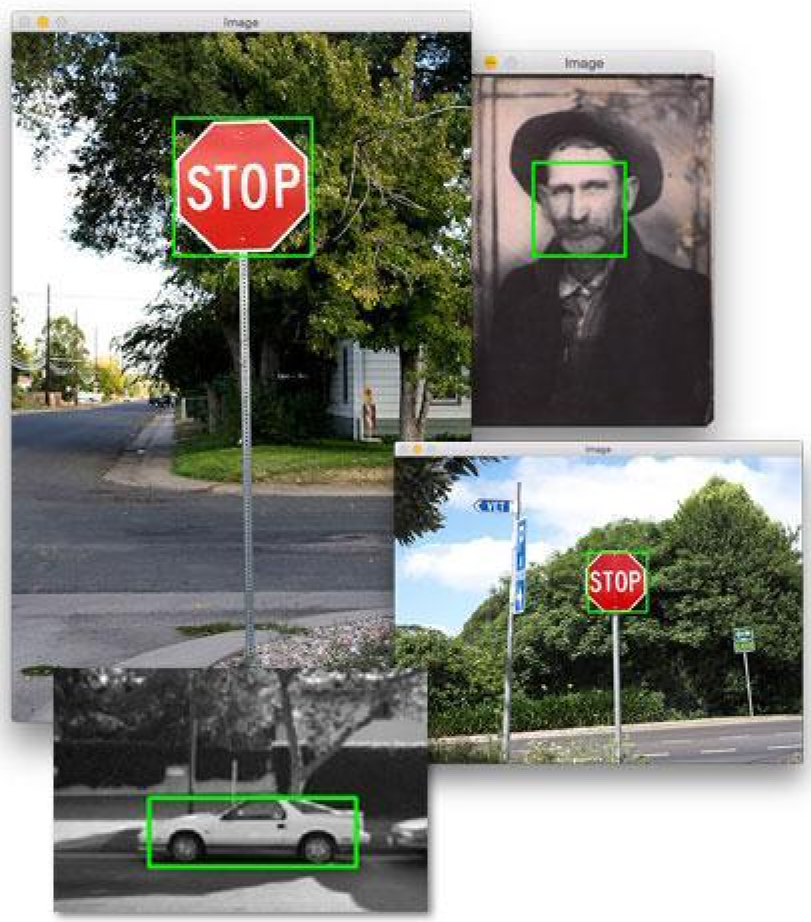
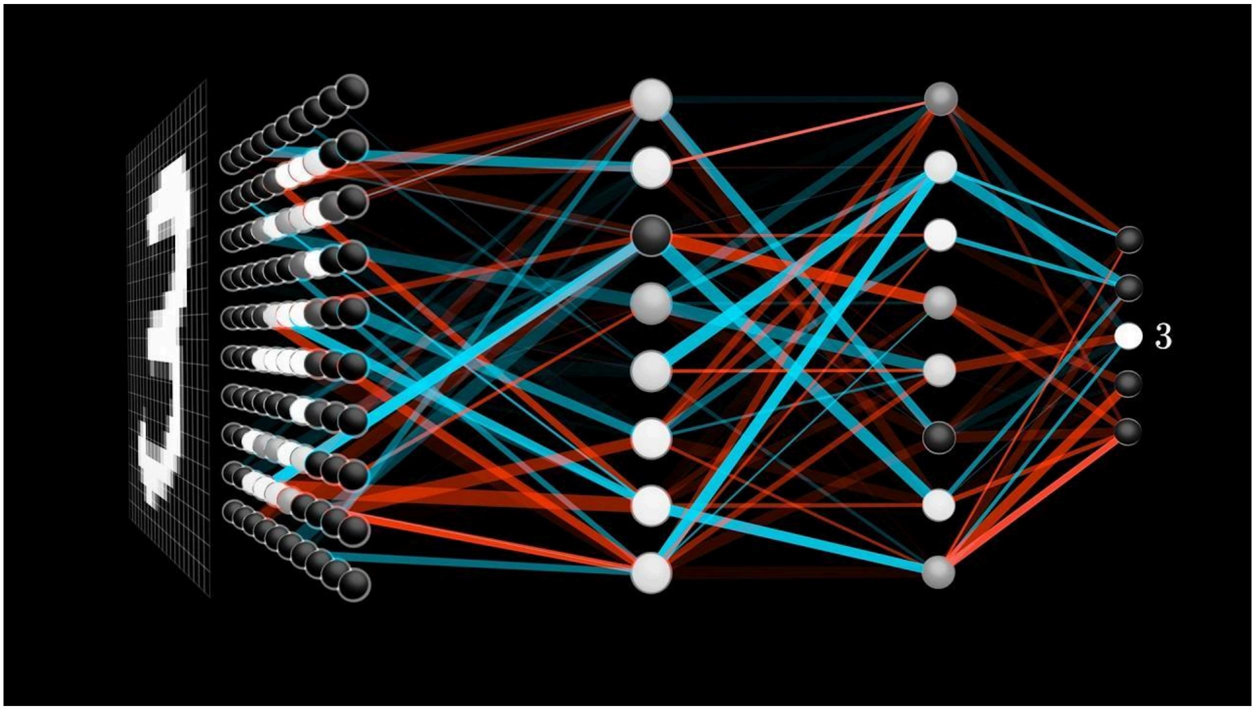


Image
Image
Image
STOP
STOP



A **Computer Vision Engineer** builds systems that can understand images and videos, such as face recognition, object detection, autonomous robots, smart CCTV, industrial inspection, and medical imaging.

Stage 1: Python Fundamentals

Learn

- Variables
- Loops
- Functions
- OOP
- NumPy
- Pandas
- Matplotlib

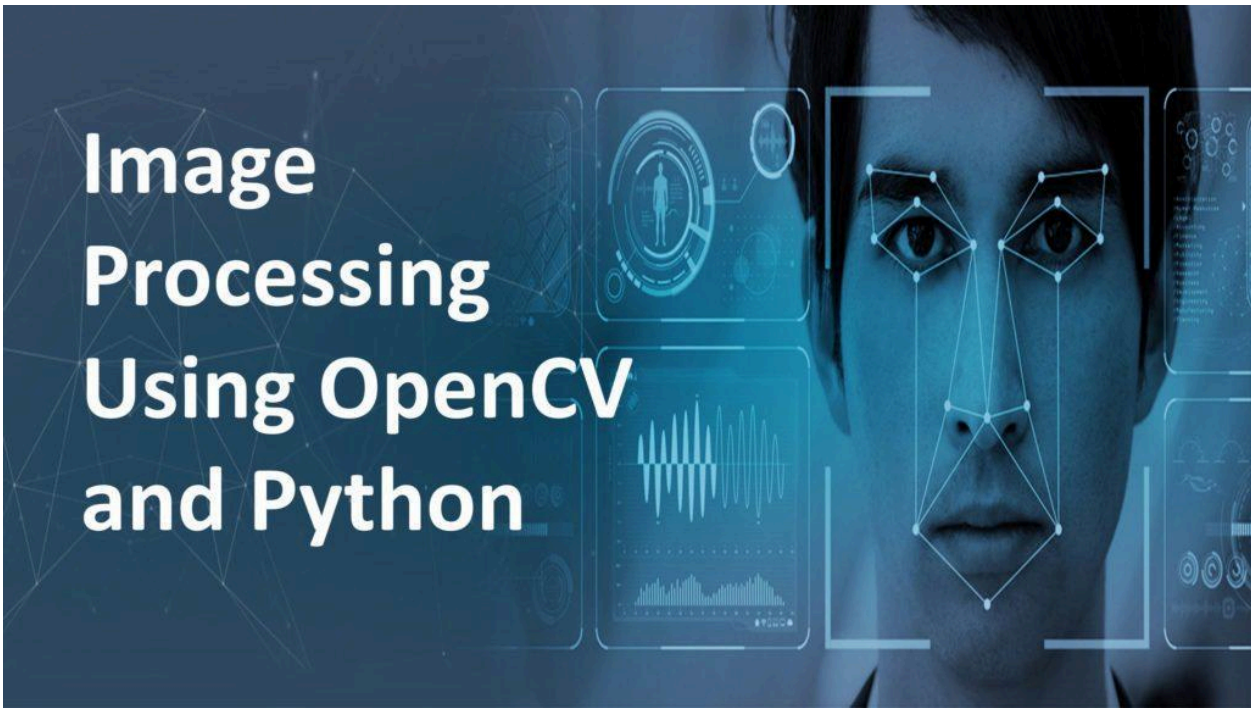
Install

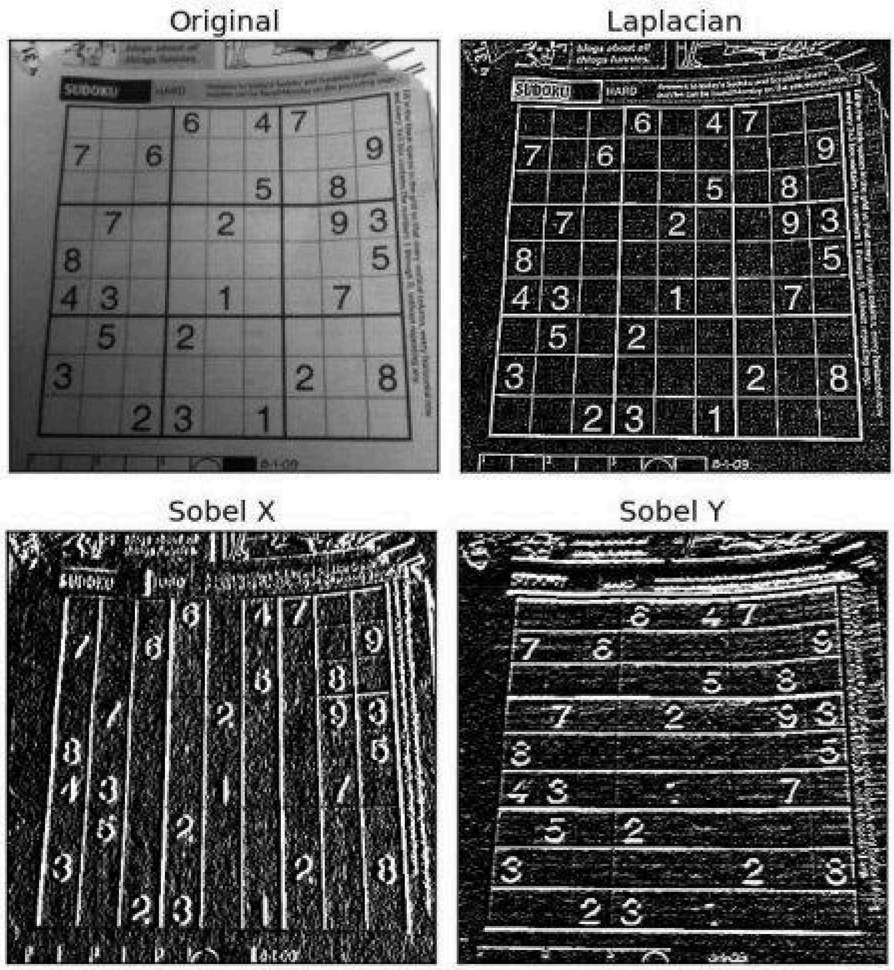
pip install numpy pandas matplotlib

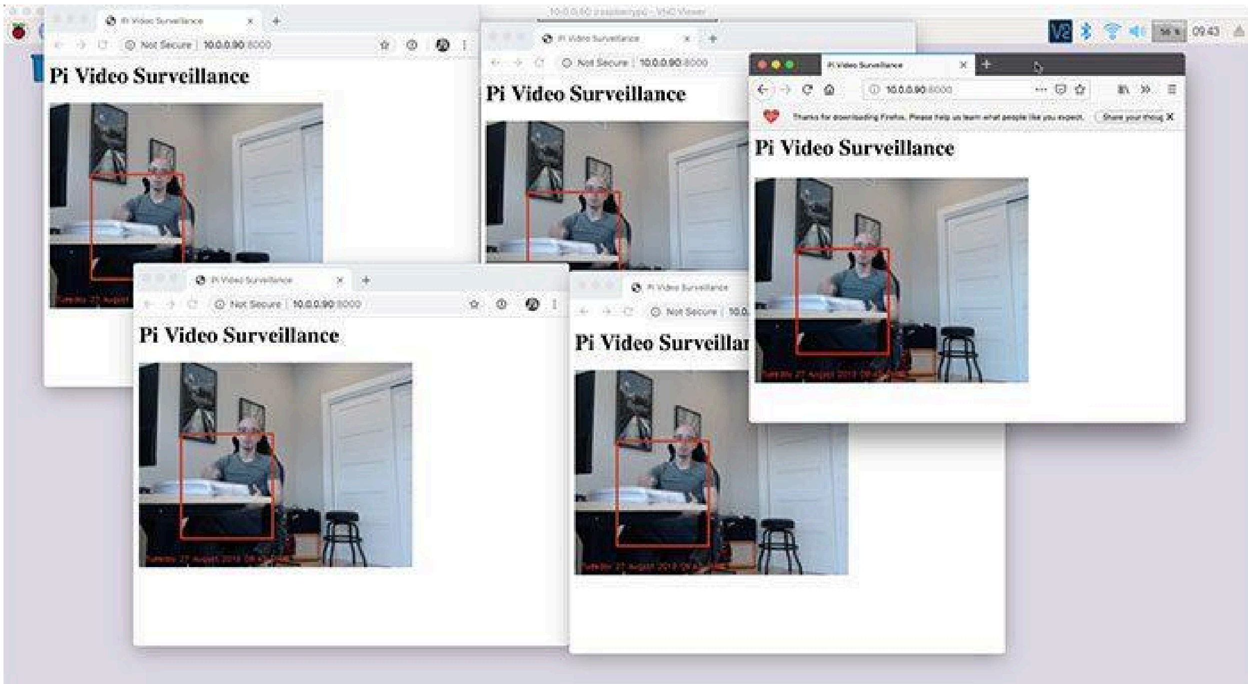
Mini Projects

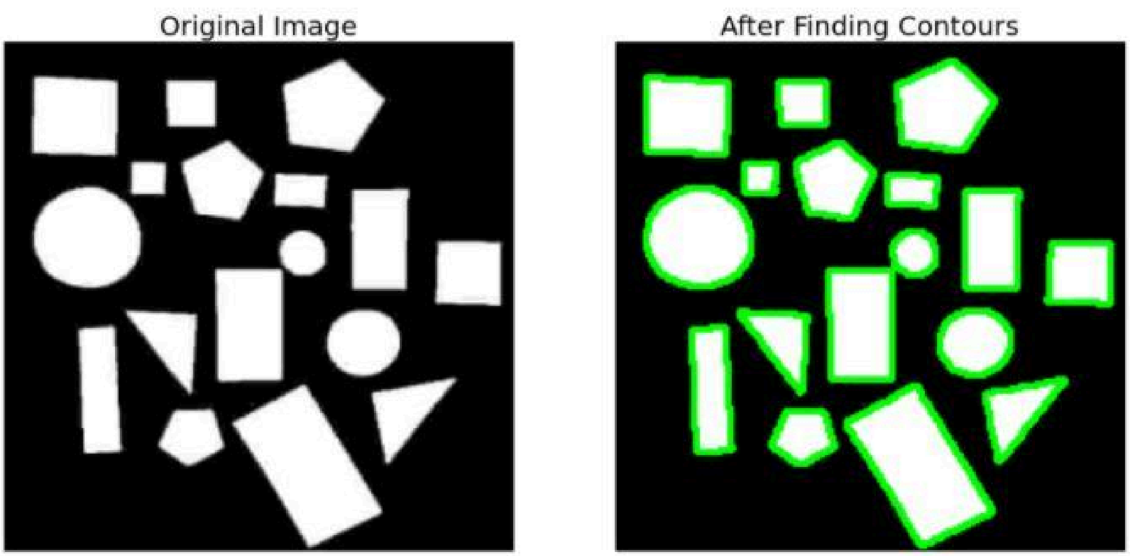
- Image Viewer
- Photo Editor
- Data Visualization

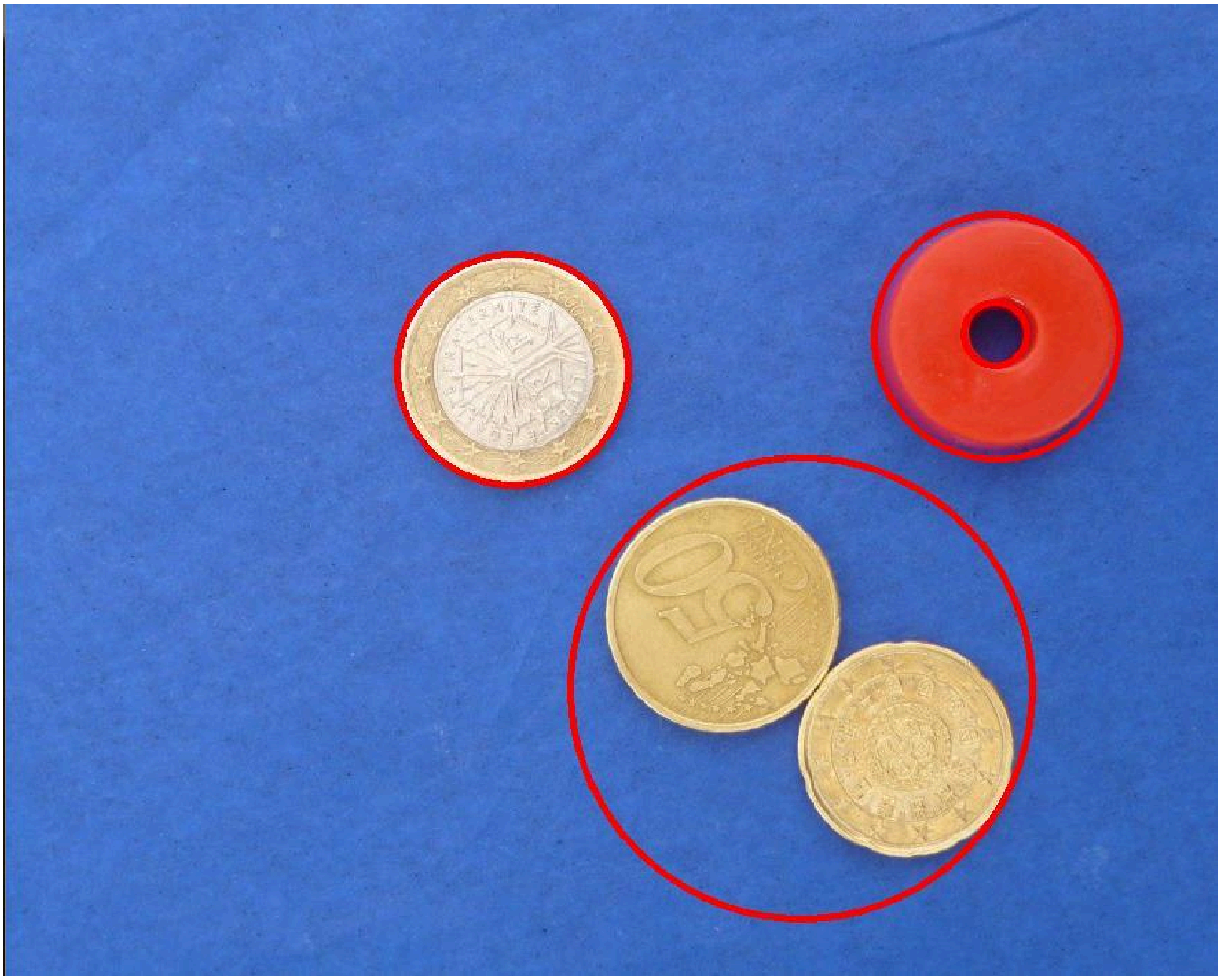
Stage 2: OpenCV

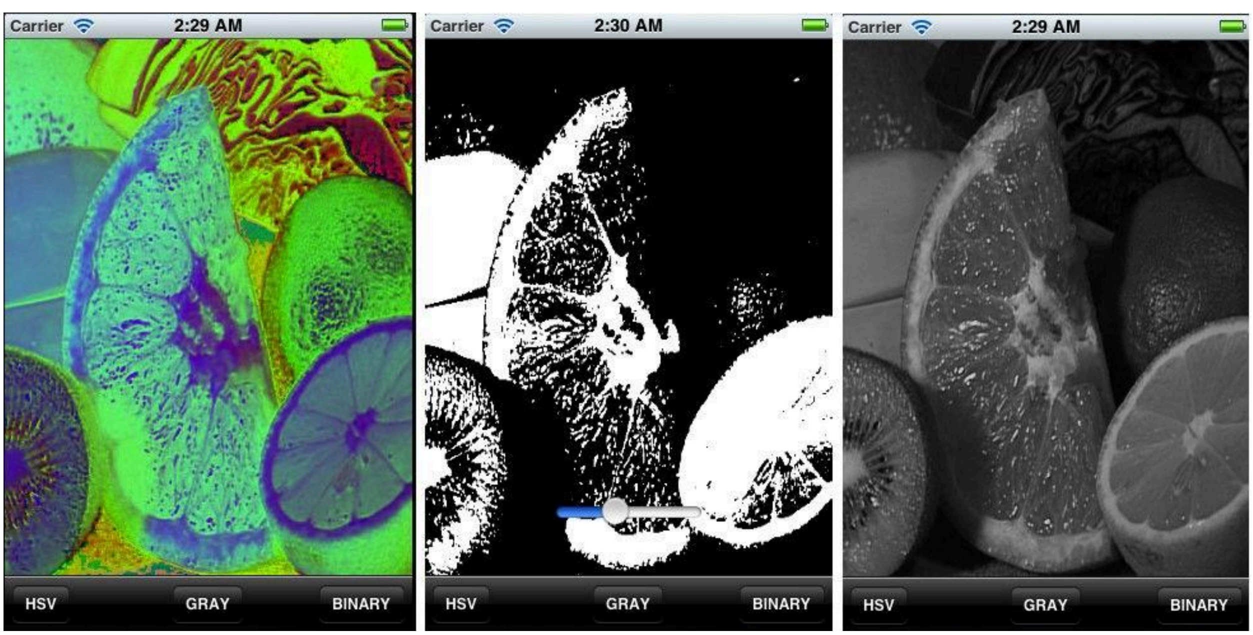



Original
Laplacian
Sobel X
Sobel Y


Pi Video Surveillance
Pi Video Surveillance
Pi Video Surveillance
Pi Video Surveillance
Pi Video Surveill


Original Image
After Finding Contours




Carrier
2:29 AM
HSV
GRAY
BINARY
Carrier
2:30 AM
HSV
GRAY
BINARY
Carrier
2:29 AM
HSV
GRAY
BINARY

Install

pip install opencv-python

Learn

Image Processing

- Read Image
- Resize
- Crop
- Rotate

Filters

- Blur
- Gaussian Blur
- Thresholding

Feature Detection

- Edges
- Contours
- Corners

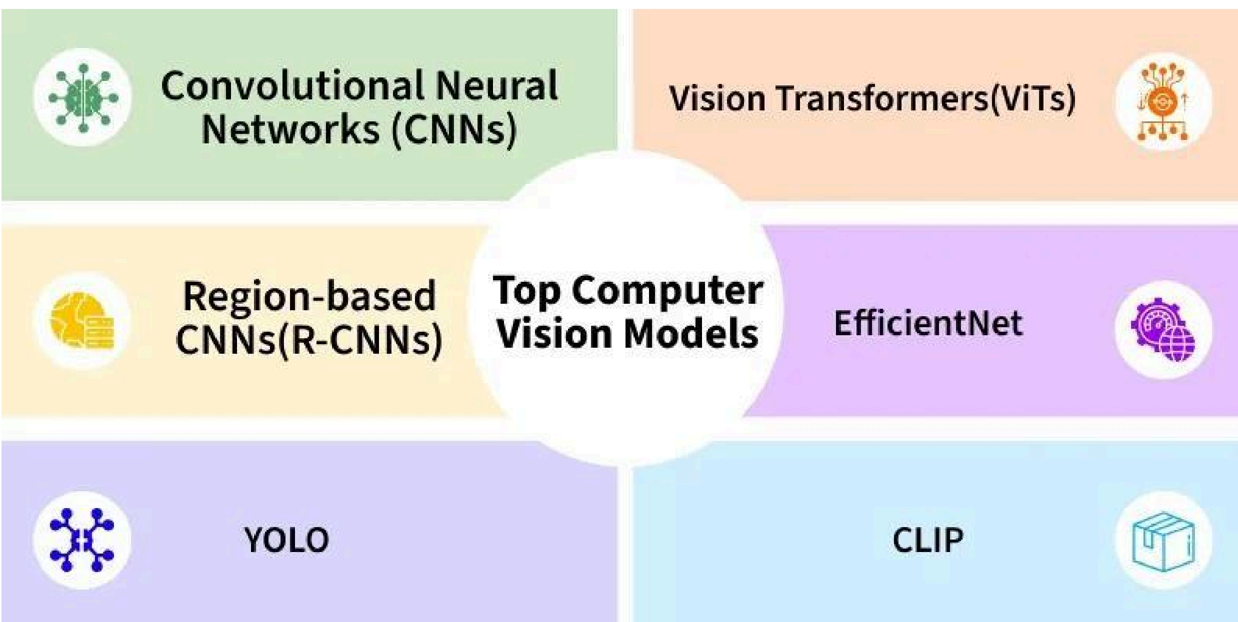
Video Processing

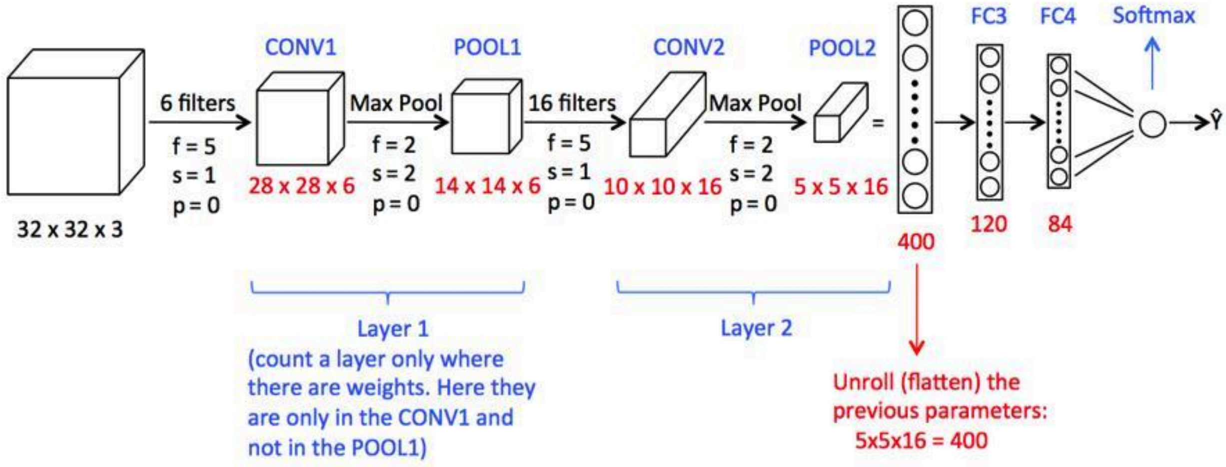
- Webcam Access
- Video Recording
- Motion Detection

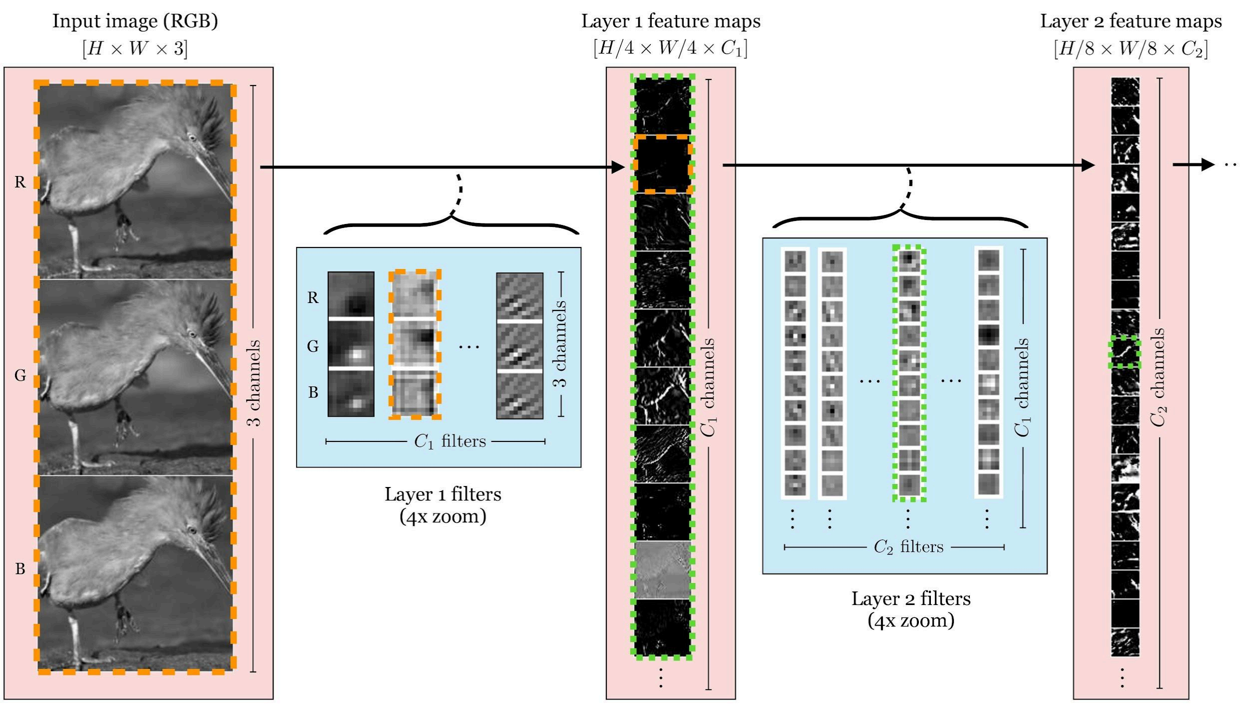
Projects

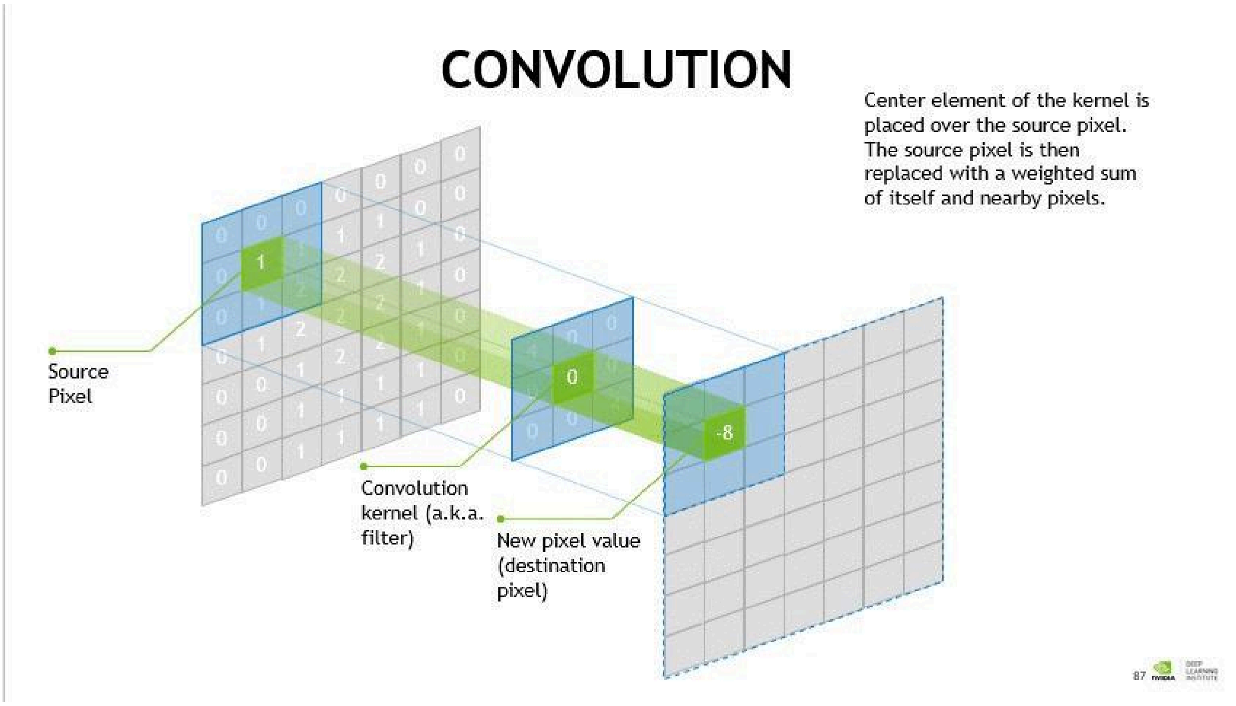
1. Motion Detection
2. Vehicle Counter
3. Lane Detection
4. Number Plate Detection
5. Smart CCTV

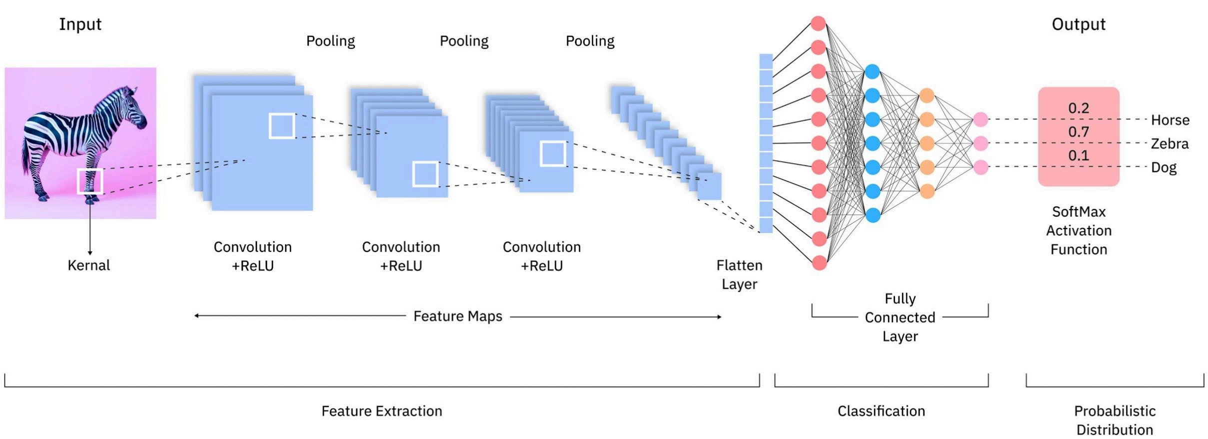
Stage 3: CNN (Convolutional Neural Networks)

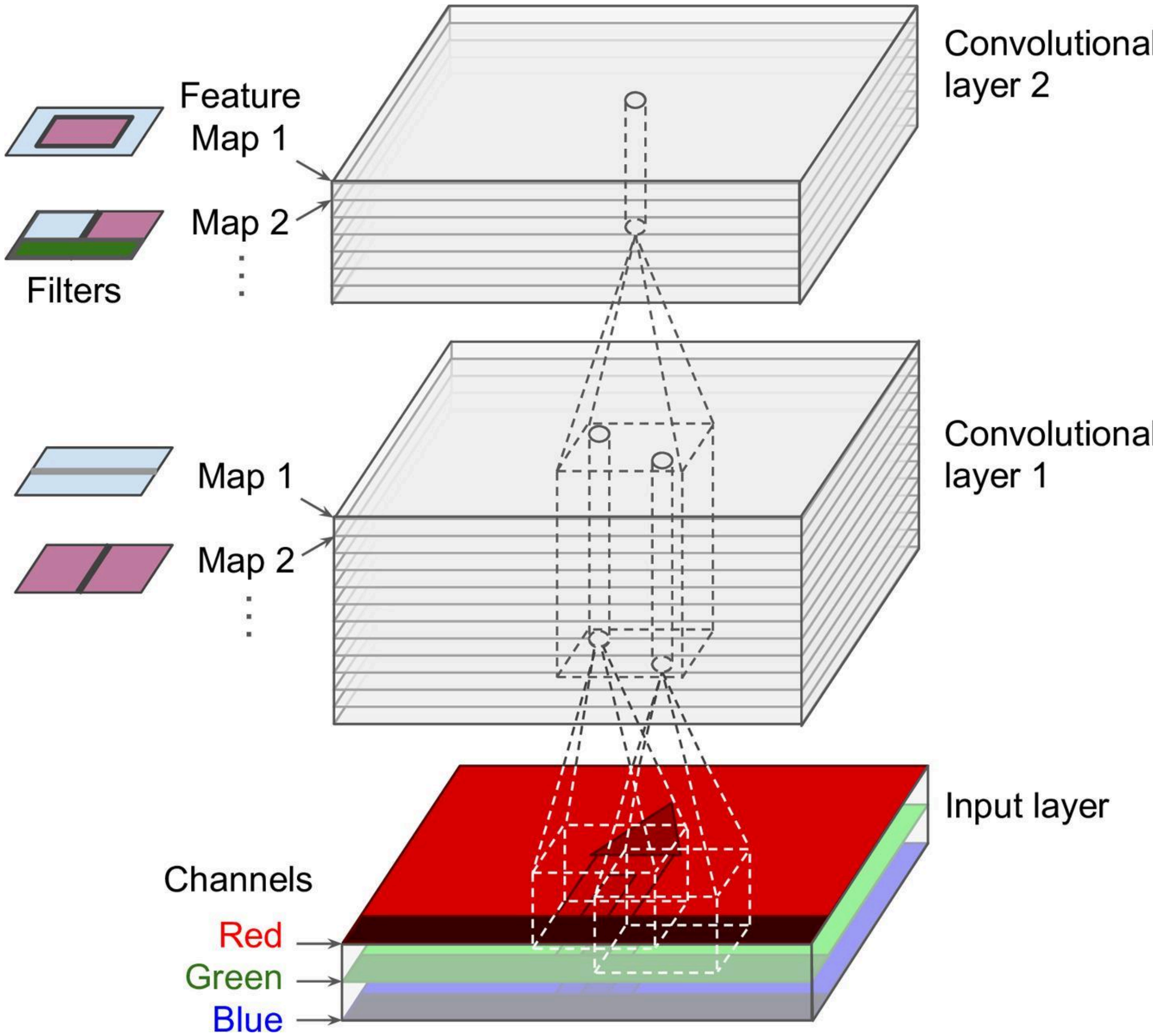



CONV1
POOL1
CONV2
POOL2
FC3
FC4
Softmax
6 filters
f = 5
s = 1
p = 0
Max Pool
f = 2
s = 2
p = 0
16 filters
f = 5
s = 1
p = 0
Max Pool
f = 2
s = 2
p = 0
32 x 32 x 3
28 x 28 x 6
14 x 14 x 6
10 x 10 x 16
5 x 5 x 16
400
120
84
Ŷ
Layer 1
(count a layer only where there are weights. Here they are only in the CONV1 and not in the POOL1)
Layer 2
Unroll (flatten) the previous parameters: 5x5x16 = 400


Input image (RGB)
$[H \times W \times 3]$
R
G
B
3 channels
Layer 1 feature maps
$[H/4 \times W/4 \times C_1]$
C_1 channels
Layer 2 feature maps
$[H/8 \times W/8 \times C_2]$
C_2 channels
3 channels
C_1 filters
Layer 1 filters
(4x zoom)
C_1 channels
C_2 filters
Layer 2 filters
(4x zoom)


CONVOLUTION
Center element of the kernel is placed over the source pixel. The source pixel is then replaced with a weighted sum of itself and nearby pixels.
Source Pixel
Convolution kernel (a.k.a. filter)
New pixel value (destination pixel)
-8
87


Input
Pooling
Pooling
Pooling
Output
0.2
0.7
0.1
Horse
Zebra
Dog
SoftMax Activation Function
Kernal
Convolution +ReLU
Convolution +ReLU
Convolution +ReLU
Flatten Layer
Fully Connected Layer
Feature Maps
Feature Extraction
Classification
Probabilistic Distribution



Install

pip install tensorflow

Learn

CNN Concepts

- Convolution Layer
- Pooling Layer
- Flatten Layer
- Dense Layer

Image Classification

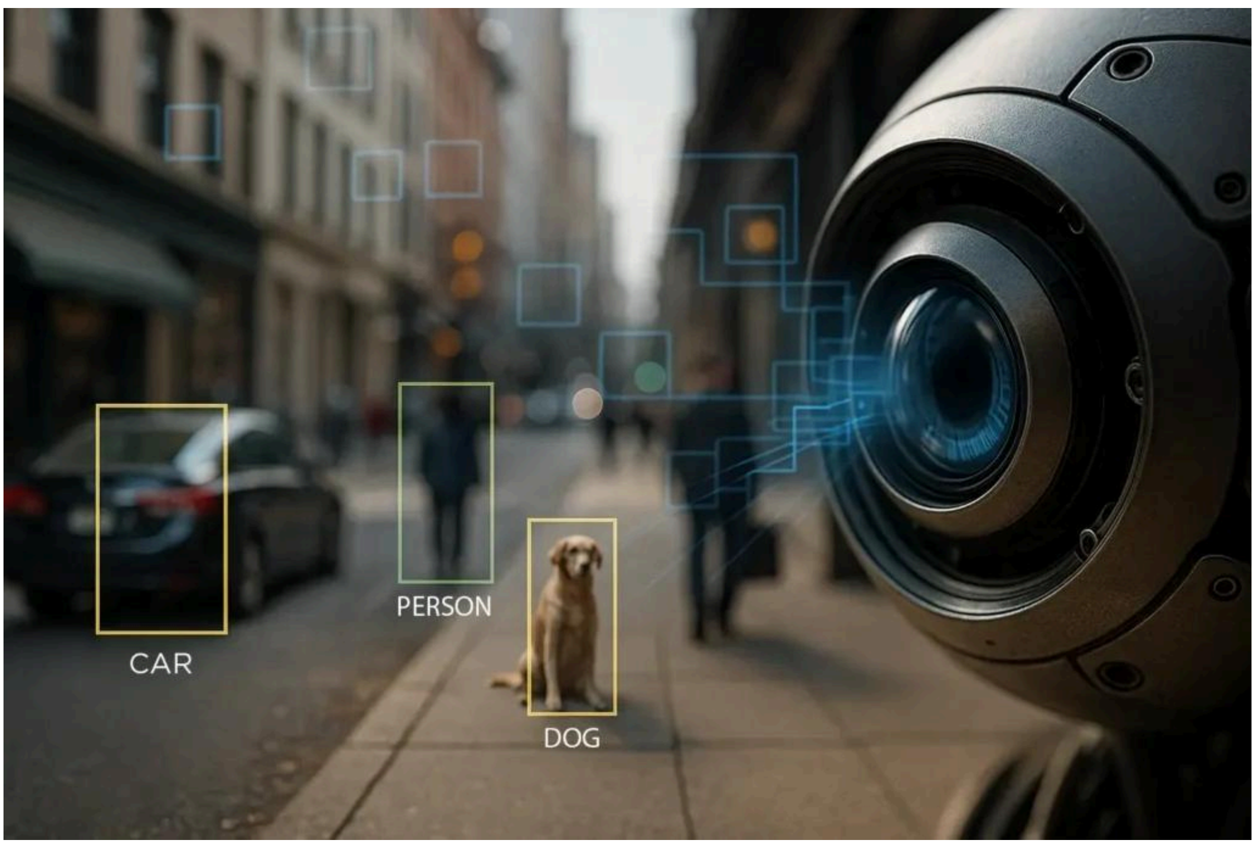
Examples:

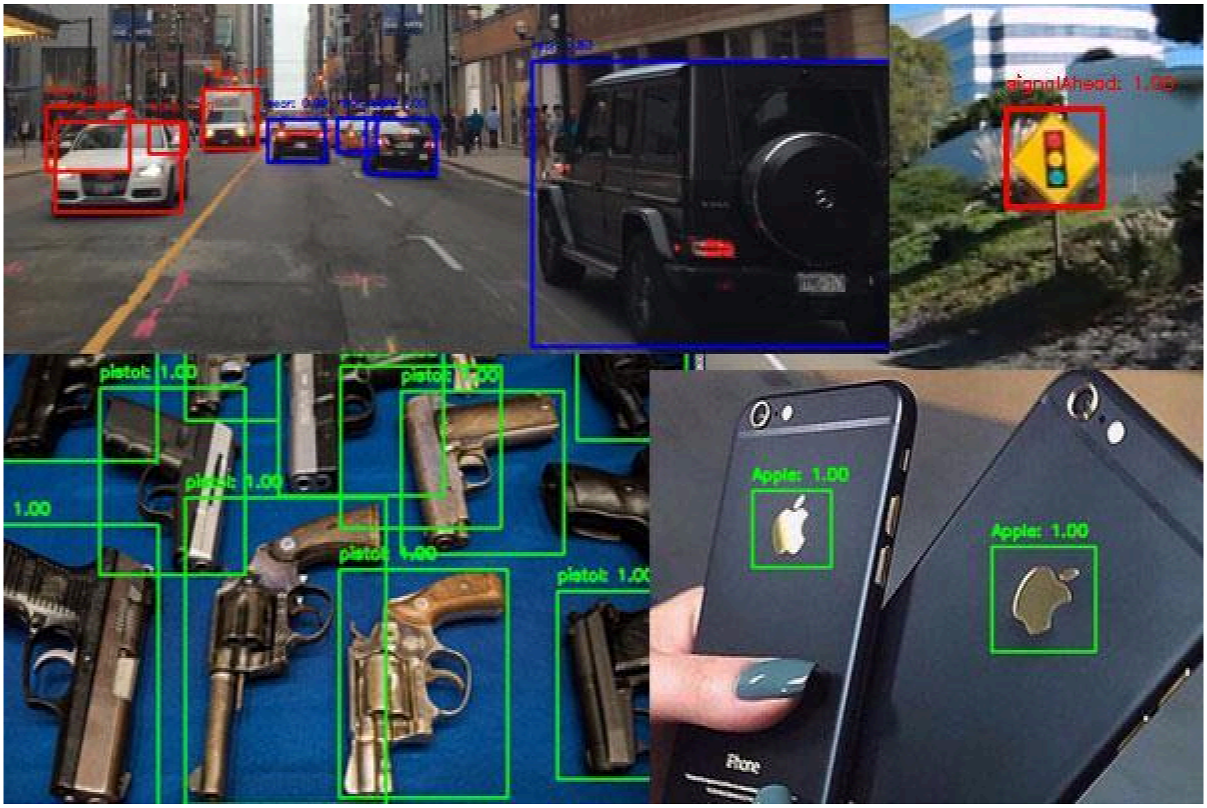
- Cat vs Dog
- Plant Disease
- Face Mask Detection

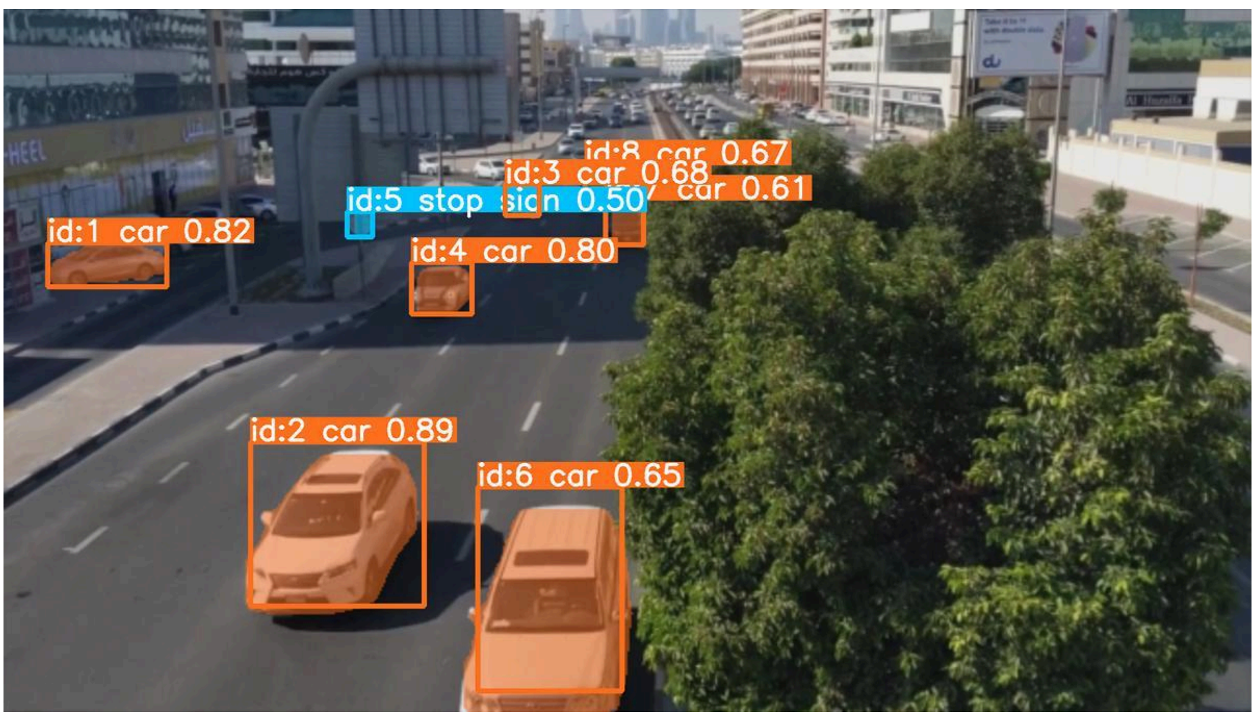
Projects

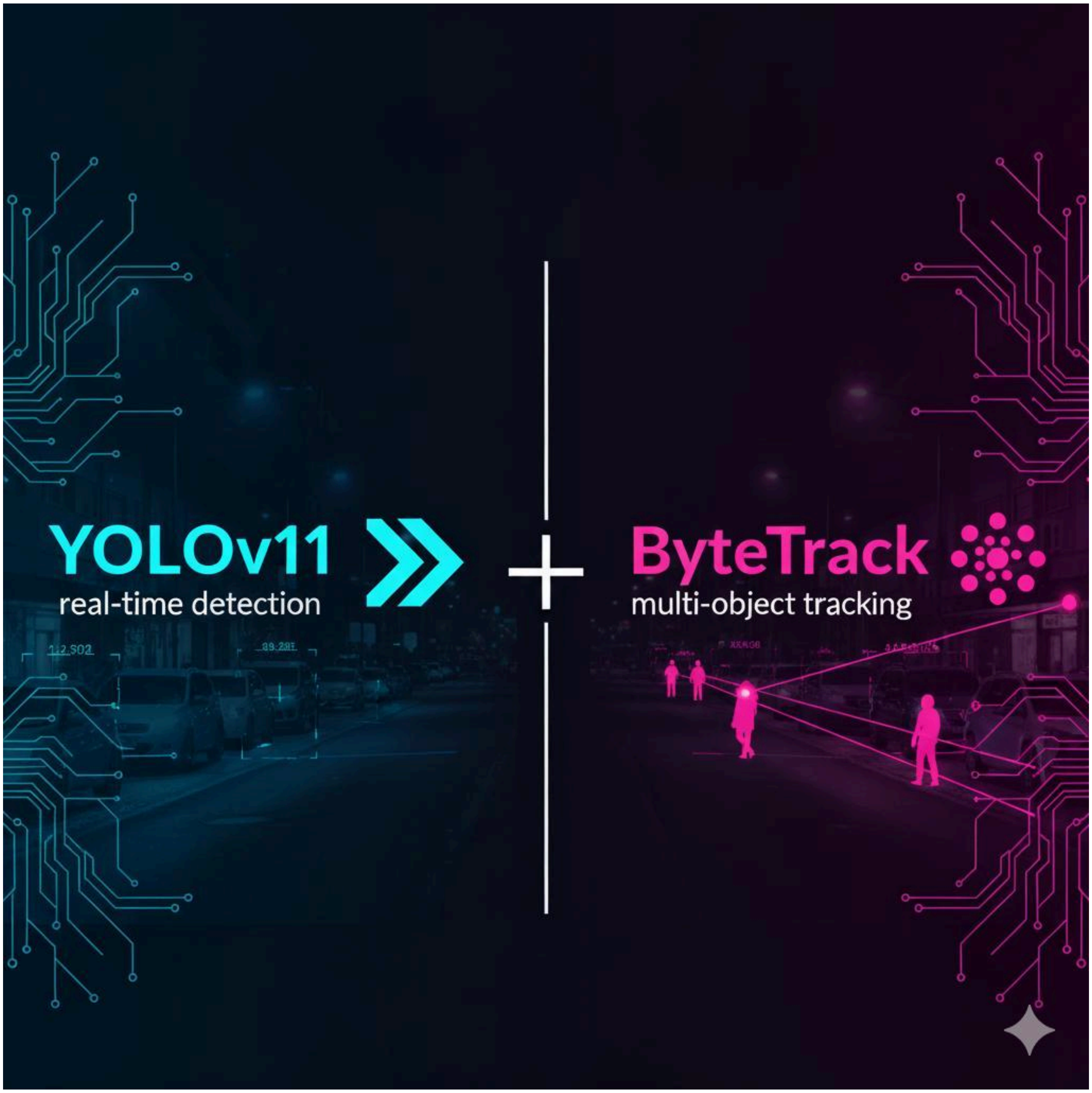
1. Face Mask Detector
2. Animal Classifier
3. Plant Disease Detection
4. Medical X-Ray Classification

Stage 4: YOLO Object Detection




signalAhead: 1.00
pistol: 1.00
pistol: 1.00
pistol: 1.00
pistol: 1.00
pistol: 1.00
1.00
Apple: 1.00
Apple: 1.00
iPhone


id:8 car 0.67
id:3 car 0.68
car 0.61
id:1 car 0.82
id:4 car 0.80
id:2 car 0.89
id:6 car 0.65

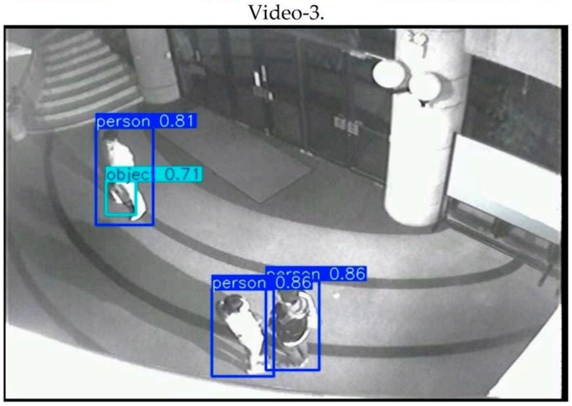

YOLOv11
real-time detection
ByteTrack
multi-object tracking

Video-1.

Video-2.

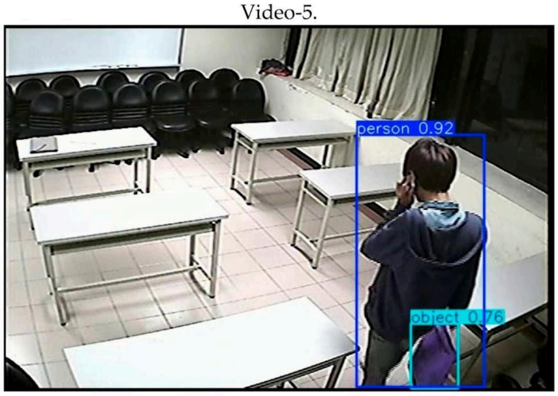
Video-3.

Video-4.

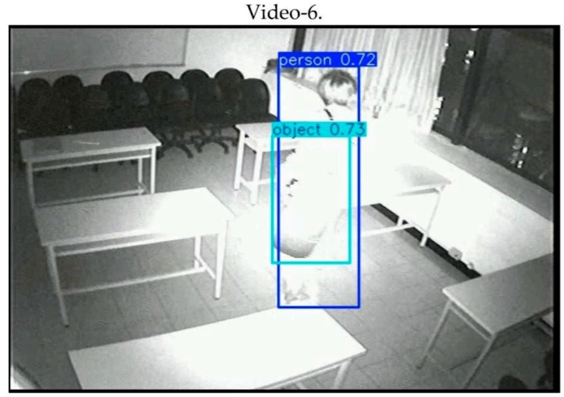


Video-5.

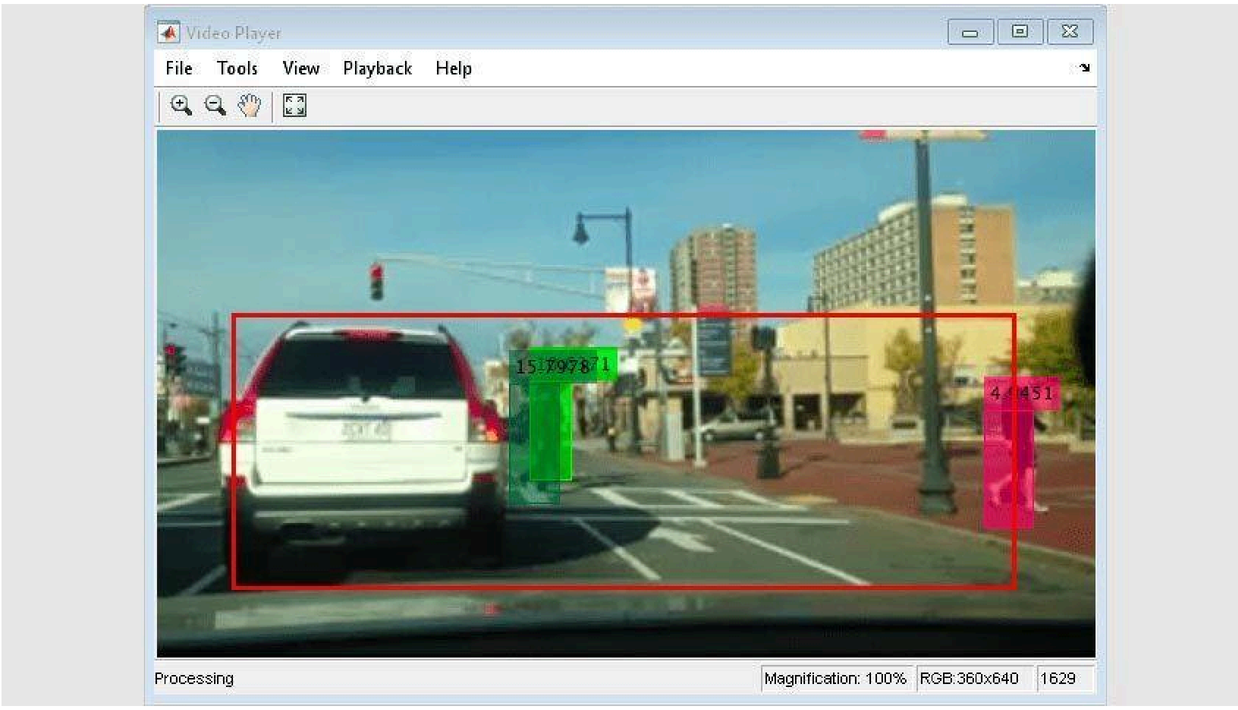
Video-6.

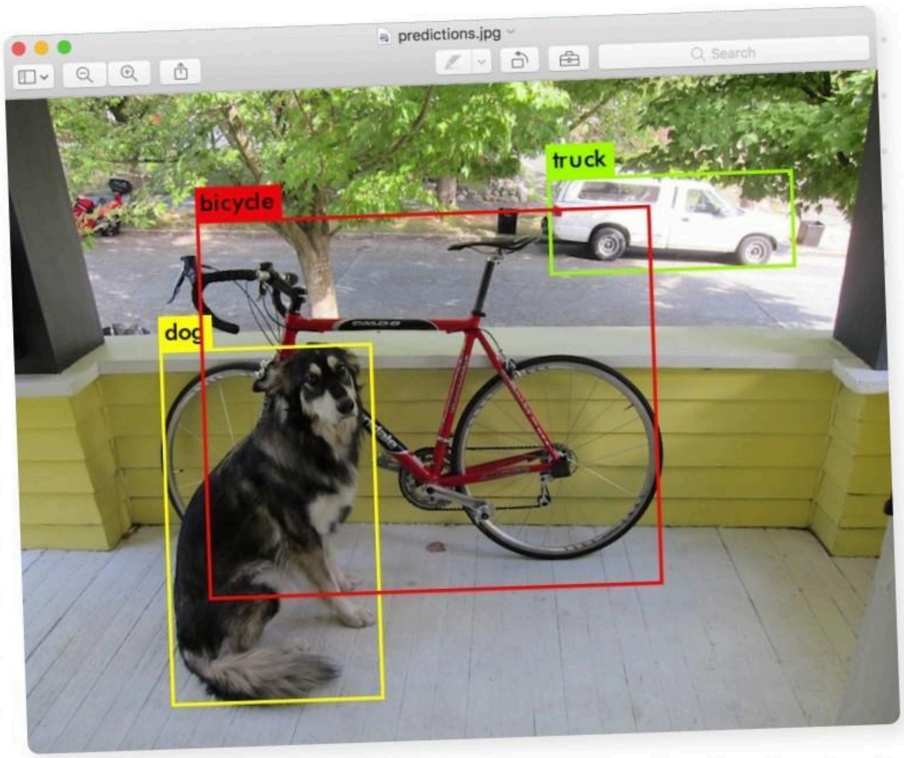


Video-7.



Video-8.


Video Player
File
Tools
View
Playback
Help
Processing
Magnification: 100%
RGB:360x640
1629



Install

```
pip install ultralytics
```

First Program

```
from ultralytics import YOLO

model = YOLO("yolov8n.pt")

results = model.predict(
    source=0,
    show=True
)
```

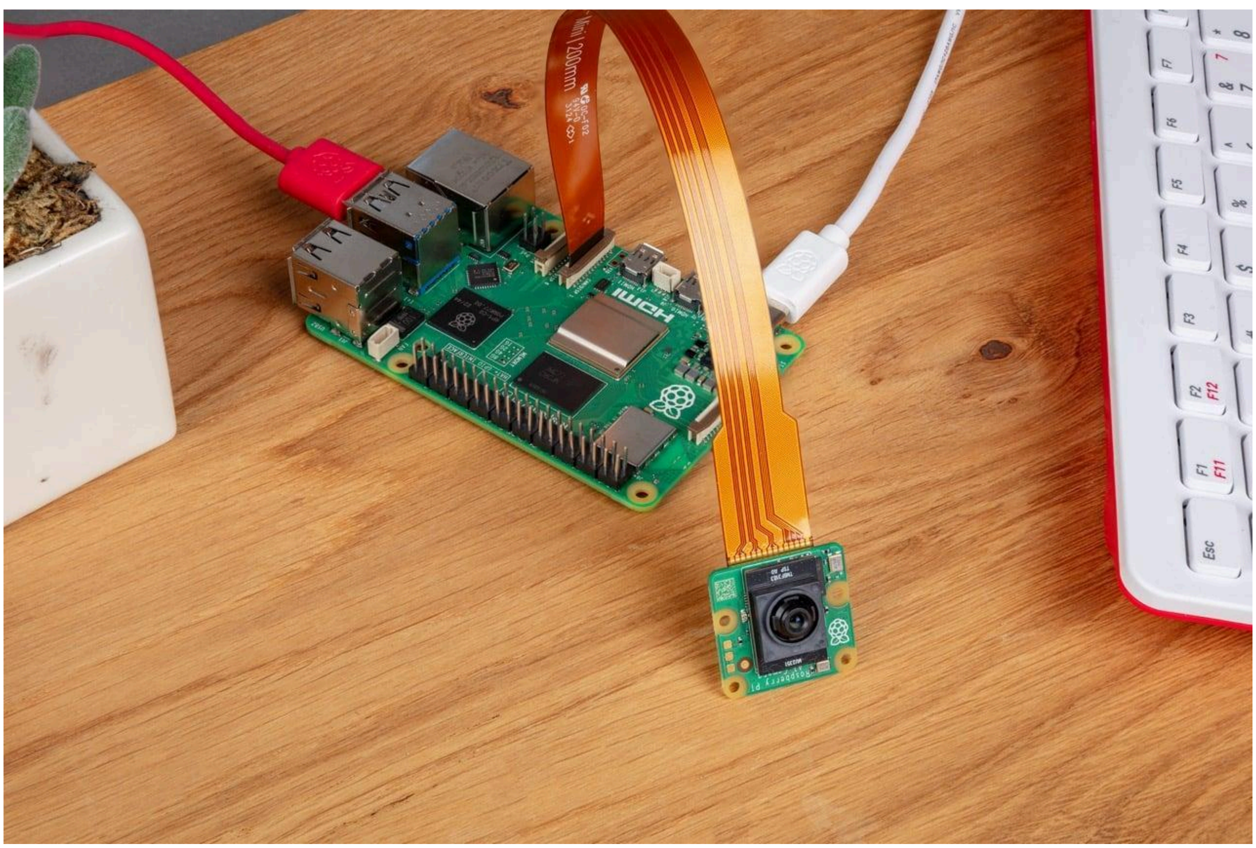
Learn

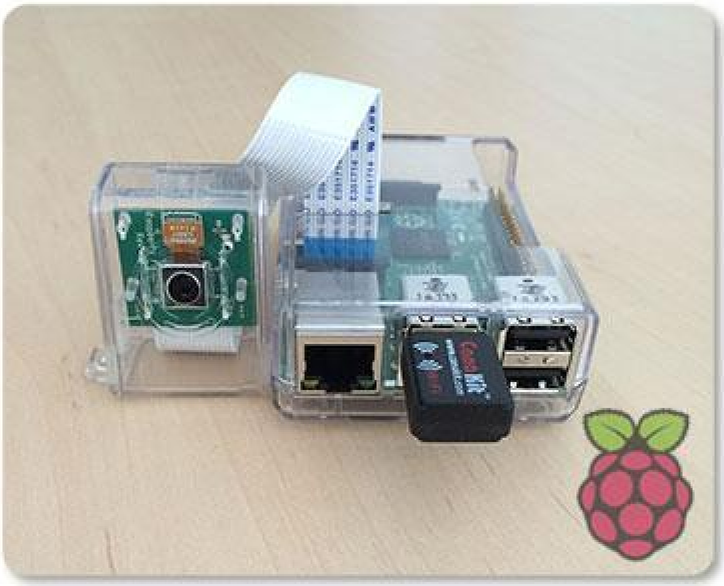
- Bounding Boxes
- Object Tracking
- Segmentation
- Pose Estimation

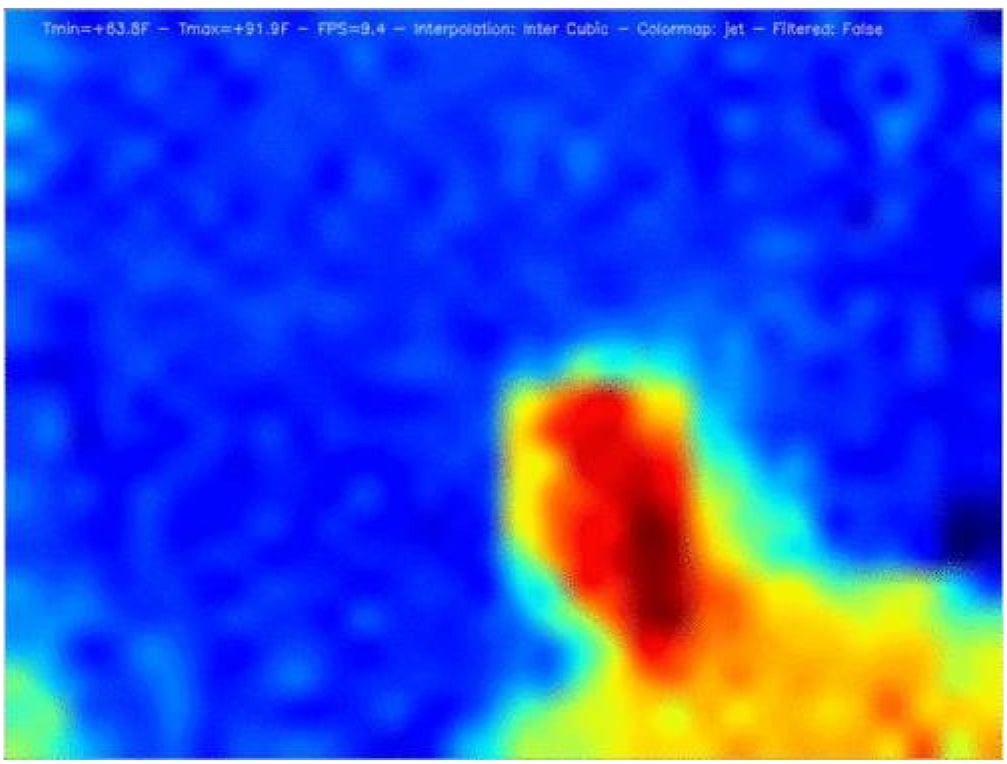
Projects

1. Human Detection
2. Vehicle Detection
3. Traffic Monitoring
4. Drone Surveillance
5. Smart Security Camera

Raspberry Pi 5 Computer Vision Projects






Tmin=+63.8F – Tmax=+91.9F – FPS=9.4 – Interpolation: Inter Cubic – Colormap: jet – Filtered: False

Hardware

- Raspberry Pi 5
- USB Camera / Pi Camera
- MLX90640 Thermal Camera
- SX1262 LoRa Module

Projects

Beginner

- Motion Detection Camera
- Face Detection Camera

Intermediate

- Face Recognition Attendance
- Vehicle Counting System

Advanced

- Thermal Human Detection
- AI CCTV Surveillance
- Smart Wildlife Monitoring
- LoRa-Based Remote Monitoring